

PRINCE OMUYEH

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EDUCATION

Villanova University

Master of Science in Software Engineering

Villanova, PA

August 2026 – Present

Lehigh University, P.C. Rossin College of Engineering

Bachelor of Science in Computer Science, Minor in Data Science

Bethlehem, PA

August 2022 - May 2026

Relevant Coursework: Data Structures & Algorithms, Operating Systems Design, Distributed Systems, Database Systems, Computer Architecture, Systems Software, Statistical Computing, AI Theory & Practice, Data Mining

SKILLS

Languages: Python, Java, C++, SQL, JavaScript, R, Rust, HTML/CSS

Frameworks & Libraries: React, Spring, Maven, Node.js, Supabase

Developer Tools: Git, Docker, Linux, Jira/Atlassian, Agile/Scrum, PostgreSQL, Kafka

EXPERIENCE

Computer Vision Engineer

Hawkeye (Capstone Sponsor)

January 2025 – December 2025

Bethlehem, PA

- Improved a real-time Python computer vision pipeline for cyclist hazard detection, collaborating with a startup sponsor on a 4-person Agile team via Jira
- Migrated the object detection model from YOLOv8 to YOLOv11 by updating model architecture configs, refactoring preprocessing and postprocessing logic, and resolving dependency changes across the Ultralytics codebase
- Achieved a vehicle orientation prediction accuracy of 92.66%

Software Development Intern

Lehigh University

March 2025 – May 2025

Bethlehem, PA

- Built an admin interface for a full-stack academic integrity platform using JavaScript and React, enabling 100+ faculty and associate chairs to manage user roles and control access to student reports
- Implemented 6 RESTful APIs and OAuth authentication supporting role-based access control for faculty, associate chairs, and student accounts

PROJECTS

Movie Watchlist Web Application | *React, Node.js, PostgreSQL*

April 2026 – May 2026

- Built a full-stack movie watchlist application with a team of 3, featuring movie search, personal watchlists, and social reviews powered by the TMDb API and a PostgreSQL database
- Implemented a complete authentication system with email/password login, Google OAuth, and JWT session management using Supabase Auth

Distributed Replicated Hash Table | *C++*

April 2026

- Implemented a distributed hash table across 5 nodes with Paxos-based consensus, supporting atomic operations, data replication, and fault tolerance
- Achieved 30,000 ops/sec throughput and sub-1ms average latency at 10k keys under fault injection testing, maintaining availability after node crashes with no manual intervention

Scripting Language Interpreter | *Rust*

April 2025

- Built a full compiler pipeline in Rust — hand-rolled lexer, recursive descent parser using nom combinators, and tree-walking interpreter — producing a working custom scripting language from scratch
- Implemented 15+ AST node types across 3 primitive types with stack-frame-based scoping for function calls

JPMorgan Chase Transaction Processing System | *Java, Spring, Kafka, SQL*

April 2025

- Designed a Kafka-based transaction processing system in Spring Boot backed by an H2 SQL database
- Developed transaction pipelines, external API interactions, and user balance endpoints using Java, Maven, and Spring